

Reminders

Upcoming due dates

TODAY: A4

Teaching evaluations for extra credit!

Next Wed Final project*, video*, team eval survey, post-course survey

Your DS future

Data Science in Practice

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Courses in DS and ML at UCSD

Did you know about the reciprocal agreement between DS and COGS??

- DS [My list of '20-21 ML \(and ML adjacent\) courses](#)
- CSE
- CS
- ECE
- COGS
- But also many other departments like ECON, MATH, LING, BENG, etc

Some job titles and what they do

- Analytics or statistician: data handling, analysis
- Data scientist: programming, data handling, analysis
- Data engineer: programming, databases, management
- Data architect: programming, databases, design
- Data manager: databases, design, management
- *OPs (eg, devOPs, dataOPs, full stack): programming, tool development, management concentrating on end to end process
- ML Engineer: programming, tool development, management of infrastructure
- ML researcher: programming, algorithm design and testing

New Roles and Job Responsibilities in 2024 from Burtchworks DS report
AI Ethics Officer: Develops, implements, and oversees ethical guidelines and practices for AI and machine learning systems. Ensures that AI applications are transparent, fair, and free from biases. Machine Learning Operations (MLOps) Engineer: Focuses on deploying, monitoring, and maintaining machine learning models in production. Collaborates with data scientists and IT teams to automate and streamline ML workflows. Data Science Product Manager: Manages the development and launch of data-driven products. Works with data scientists, engineers, and stakeholders to ensure product alignment with business goals and user needs. AI Research Scientist: Conducts cutting-edge research in artificial intelligence, focusing on developing new algorithms, models, and techniques. Publishes findings in scientific journals and presents at conferences. Deep Learning Engineer: Specializes in building and optimizing deep neural networks for tasks such as image recognition, natural language processing, and speech recognition. Works with large datasets and high-performance computing resources. Natural Language Processing (NLP) Engineer: Develops and implements algorithms for understanding and generating human language. Works on applications like chatbots, sentiment analysis, and language translation. AI Solution Architect: Designs and implements AI solutions to solve complex business problems. Works closely with stakeholders to understand requirements and develop scalable AI architectures. Data Privacy Officer: Ensures that data handling practices comply with privacy laws and regulations. Works on protecting sensitive information and developing data governance policies. Computer Vision Engineer: Focuses on developing algorithms and systems for analyzing and interpreting visual data from the real world. Works on applications like autonomous vehicles, facial recognition, and medical imaging. AI Educator/Trainer: Provides training and education on AI and machine learning topics. Develops curriculum and delivers courses, workshops, and seminars to students, professionals, and organizations. Robotics Engineer: Designs, builds, and programs robots and robotic systems. Works on integrating AI and machine learning to enhance robotic capabilities and autonomy. Data Storyteller: Translates complex data analysis into compelling narratives and visualizations. Communicates insights to stakeholders in a clear and impactful manner. Cognitive AI Developer: Develops AI systems that mimic human thought processes. Works on applications such as decision support systems, cognitive automation, and AI-based problem-solving tools. Federated Learning Engineer: Specializes in federated learning, where machine learning models are trained across multiple decentralized devices or servers. Focuses on privacy-preserving and collaborative AI. AI Quality Assurance (QA) Specialist: Ensures the quality and performance of AI models and systems. Conducts testing, validation, and evaluation to identify and resolve issues before deployment. Autonomous Systems Engineer: Develops and maintains autonomous systems, such as self-driving cars or drones. Integrates AI algorithms to enable autonomous decision-making and operation. AI Personalization Specialist: Focuses on creating personalized user experiences using AI. Works on recommendation systems, adaptive learning platforms, and personalized marketing. AI-driven Healthcare Specialist: Develops AI applications for the healthcare industry. Works on predictive analytics, medical imaging analysis, and personalized treatment plans. AI Regulation and Compliance Specialist: Ensures that AI systems comply with regulatory requirements and standards. Works on developing frameworks for legal and ethical AI use. Voice and Speech Recognition Engineer: Develops algorithms and systems for recognizing and synthesizing human speech. Works on applications like virtual assistants, transcription services, and voice biometrics.

What should you learn next?

- Data science?
 - Statistics + Modeling
 - SQL and/or Data viz packages
 - Cloud!
 - Become fantastic at one of Python/R and learn many packages. Be capable at the other language
- AI?
 - Statistics, Vector calculus, ML theory
 - Cloud!
 - Foundation models
 - Become usefully good at several frameworks (PyTorch, Go, Tensorflow, etc)
 - Read arXiv and NeurIPS/ICLR papers
- Both: learn to properly use AI tools to help you :^/

How do you use AI

How to properly use AI tools

As a programmer

- Beginner
 - Define a narrow task like “write a function to individually z-score only the numeric columns of a Pandas dataframe. The operation should not modify the data frame in place, it should generate a copy of the original data frame and return that copy “
 - Ask AI to explain errors to you
 - Ask it to suggest better approaches to a problem
 - Ask it to write a unit test, or in chat ask it to back and forth with you on what kinds of tests make sense
- Agentic approaches
 - Make a detailed English planning document in a .md file
 - Set the agents to work on different parts of it, including allowing them to modify and expand the plan
 - e.g. <https://kiro.dev/>

How to properly use AI tools

As a data scientist

- Define a narrow task like “write a function to individually z-score only the numeric columns of a Pandas dataframe. The operation should not modify the data frame in place, it should generate a copy of the original data frame and return that copy”
- Ask it to help with writing API calls to accomplish something
- Ask it to help with data cleaning or generating summary reports
- You need to supply the deep thinking and analysis... its a tool to get it done more efficiently!

2025 Burtchworks Datascience and AI Professionals Salary Report

ROLE CLASSIFICATION & SAMPLE

Burtch Works differentiates these groups by their **primary mission and deliverables**, not just data modality.

Data Scientists focus on driving decisions with statistical analysis and classical ML. They own experimentation and causal inference, forecasting, segmentation, and predictive modeling on business/operational data; their outputs are analyses, validated experiments, and models embedded in analytics/BI or batch/stream scoring.

AI Professionals design, build, and operate **ML/LLM-powered systems** that interact with users and workflows. They work across un/semi-structured and multimodal data; fine-tune and distill models; implement retrieval/grounding; orchestrate tools/agents; run RLHF & safety evaluations; and manage MLOps (serving, monitoring, cost/latency, governance).

Roles increasingly overlap; we classify each professional by what they primarily do day-to-day in their current job.

2025 Sample

724 Data Scientists and 162 AI Professionals (total **886**), based on a strict 12-month window and the same segmentation used in prior years.

Burtch Works has tracked Data Science and AI compensation for over a decade. The 2025 cycle (May 2024 – April 2025; **724 Data Scientists** and **162 AI Professionals**) shows a market that's moved from experimentation to **operational AI** — and pay is adjusting to where the scarcest, most business-critical skills now sit.





Where we are
TODAY

AI & Data Science Hiring Landscape

FROM HYPE TO HANDS-ON

Gen-AI moves from pilots to production
Hiring shifts from “prompt roles” to **Applied LLM Engineers** who can integrate, monitor, and cost-optimize real workflows.

RLHF & alignment stay supply-constrained (+34% reqs vs +7% supply)
Scarcity drives premiums and pushes teams toward train-to-hire pipelines and selective offshore/nearshore augmentation.

AI Innovation Labs
Labs are evolving into **productization engines** — owning reference architectures, governance, and sprint pods that turn POCs into shipped, ROI-tracked features.

Toughest new-grad market in 10 years
Fewer research-track seats; entry roles skew to AI Operations or automation, making internships and portfolios the decisive hiring requirement.

IC-led hiring dominates
Firms favor high-leverage ICs who can ship PoCs rapidly.

Increased demand (but with more selective hiring)
AI-related job postings in the U.S. more than doubled from 66,000 to nearly 140,000 between January and April 2025. Despite the surge, economic uncertainty is prompting employers to be more selective.

Broader business value and cross-functional demand
AI and Data Science hiring is expanding beyond engineering into growth, marketing, sales, and customer success functions.

Key Takeaways

Q3/Q4 HIRING SURVEY

75%

of new hires are for **Data & Analytics** roles. Market Research, Engineering, and Cyber Security also reflect current enterprise priorities.

60%

of companies are **holding steady** with their current teams, indicating a conservative hiring posture, while nearly **20%** are downsizing.

68%

of new hires are for **full-time roles**, showing a commitment to long-term workforce development.

60%

of new hires are for **backfills**, suggesting internal attrition or restructuring.

Small-scale hiring

of 1-5 roles is the norm, with very few large hiring initiatives.

Technical skills

are the top priority, followed by critical thinking, both outranking cultural fit and industry familiarity.



Hiring Demand
OVERVIEW

AI Matures, Delivers Value

AI MOVES FROM EXPERIMENTS TO SCALABLE PLATFORMS

From pilots to platforms
Enterprises consolidate tools and standardize on an AI platform stack, shifting budgets from POCs to **repeatable, ROI-tracked features**.

Cost-to-value discipline
Token, context, and inference costs are now product KPIs. Teams adopt **distillation, caching, PEFT**, and **small/efficient models** to hit latency and unit-economics targets.

Role evolution
“Prompt engineer” titles fade into **Applied LLM Engineer, Retrieval Engineer**, and **AI Platform Engineer**. Demand rises for **RLHF/Alignment Ops** and **AI Evaluation/Safety Specialists**.

Multimodal & agents mature
Voice/vision inputs and **tool-using agents** leave labs for service, sales, and ops — measured on containment, CSAT, and cost per resolution.

Hybrid buy-build wins
Teams combine vendor services with **open-weights** and in-house fine-tuning for control, privacy, and portability.

Top use cases
Knowledge search/RAG, copilots for sales & service, analytics summarization, and workflow automation. Low-ROI “general chatbots” decline.

Big capital flow & infrastructure buildouts
Massive investments in data center infrastructure and workforce development. Private equity is fueling hyperscalers and co-locations to meet rising demand for AI compute as global data center spending could hit \$3 trillion in 2029.

Geographical & regulatory shifts
The landmark “TAKE IT DOWN Act” criminalizes the publication of non-consensual intimate imagery (weather authentic or AI-generated). The One Big Beautiful Bill Act (BBB) allows companies to immediately deduct 100% of domestic R&D expenditures, including salaries, equipment, and related costs — a significant departure from the previous multi-year amortization requirement.

TEAM COMPOSITION & TALENT STRATEGY

Gen-AI Implementation (Applied LLM in production)

Role families & representative titles

- Applied LLM Engineer, Retrieval Engineer, Agent/Orchestration Engineer
- AI Solutions Architect (Apps), AI Product Engineer

What they do

- Ship user-facing copilots/agents and RAG apps; design tool-use/function-calling, memory, and guardrails.
- Optimize chunking, embeddings, rerankers, and grounding data quality.

Must-have skills

- Python/TypeScript; LangChain/LlamaIndex or equivalent; vector DBs; PEFT/LoRA; distillation.
- Evaluation & observability (quality, latency, cost per resolution); prompt hardening.

KPIs

- Containment %, CSAT, tasks/hour, \$ per task/token, latency p95, A/B win-rate vs baselines.

2025 market note

- Hiring shifts from “prompt engineer” titles to Applied LLM / Retrieval / Agent roles; entry-level comp up, mid-level AI bases flat/down.

Data Science & Analytics

Role families & representative titles

- Product/Decision Data Scientist, Causal Inference Scientist, Forecasting/Time-Series Scientist
- Marketing/Revenue Analytics, Fraud & Risk, Experimentation Lead

What they do

- Drive decisions and measurable lift: experimentation, causal models, forecasting, pricing, segmentation.
- Partner with product/ops to productionize models into BI, batch/stream scoring, or features.

Must-have skills

- A/B and quasi-experimental design, uplift models, time-series, feature engineering, SQL/Python.
- Communicating impact; cost/benefit and instrumentation.

KPIs

- Incremental revenue/savings, test velocity, forecast error, adoption of DS outputs.

2025 market note

- DS leadership (MG-2/MG-3) re-priced up as orgs operationalize analytics and govern AI programs.

TEAM COMPOSITION & TALENT STRATEGY

AI-Lab Build-Out

Role families & representative titles

- Head of AI / Lab Director (Director/VP), AI Product Lead, Technical Program Manager (AI)
- AI Solutions Architect (LLM/RAG), Governance & Risk Lead (MRM), Data Sourcing Lead
- Evaluation/Safety Lead, Labeling/Ops Manager

What they do (2025)

- Stand up a reference architecture (data prep → retrieval → serving → eval) and an operating model (charter, intake, governance).
- Prioritize use cases, build the roadmap, staff pods, and own vendor/open-weights strategy.
- Institutionalize evaluation, safety, privacy, and spend controls from day one.

Must-have skills

- Translating business goals into AI roadmaps; RAG/agents patterns; platform thinking.
- Model risk management (MRM), policy, data lineage/consent; auditability.
- Org design & hiring: sequencing IC-heavy pods before adding management layers.

KPIs

- Time-to-first-POC / time-to-prod, adoption, governance pass-rates, cost per task / latency SLOs.

2025 market note

- CoE leaders with ops + governance chops command a premium; many orgs delay VP hires but will pay for a Director who can ship.

RLHF & Alignment

Role families & representative titles

- RLHF Trainer / Rater (Onshore/Nearshore), RLHF Ops Manager, Preference Model Engineer
- Safety/Policy Engineer, Red-Team Lead, Evaluation Engineer

What they do

- Design guidelines, collect preference data, train reward/preference models, and run DPO/ORPO pipelines.
- Operationalize safety: policy taxonomies, automated eval harnesses, and incident response.

Must-have skills

- Sampling/label quality control; reward modeling; safety eval (toxicity, bias, leakage).
- Tooling: label platforms, prompt/eval frameworks, LLM Guardrails, synthetic data generation.

KPIs

- Aligned-response rate, policy-violation rate, reward model accuracy, data throughput/cost.

2025 market note

- Supply-constrained domestically; teams blend onshore leads with nearshore scale to meet SLAs.

TEAM COMPOSITION & TALENT STRATEGY

MLOps & Platform Engineering

Role families & representative titles

- ML Platform Engineer, Model Reliability/SRE (AI), Data/Streaming Engineer
- Evaluation Platform Engineer, Model Governance Engineer

What they do

- Provide the paved-road: CI/CD for ML/LLM, feature stores, model registry, inference gateways.
- Production observability (quality, drift, latency, safety) and cost controls; multi-cloud portability.

Must-have skills

- Kubernetes, GPUs, Triton/Transformers inference, Ray; eval pipelines; policy & PII controls.
- Infra as code; telemetry & alerting; blue/green and canaries for models.

KPIs

- Deployment frequency, lead time, change-fail rate, MTTR; drift incidents; infra \$ per request.

2025 market note

- “Buy the platform, build the fit” hybrid is standard — engineers who stitch vendor + open-weights are prized.

Chief Data Officer (CDO)

Role families & representative titles

- Chief Data Officer, Head of Data & Analytics, SVP Data Strategy, Chief Analytics Officer, Chief Data Analytics Officer, Chief AI Officer, Chief Strategy Officer

What they do

- Set enterprise-wide data strategy aligned to business goals, including governance, analytics, automation of low value actives, new role design and AI adoption.
- Drive data monetization initiatives, advanced analytics programs, and AI/ML integration into products, operations and business process.

Must-have skills

- Advanced analytics, AI/ML strategy, and business use case identification.
- Strong leadership and change management skills to shift organizational culture toward data fluency.

KPIs

- Adoption rate of analytics/AI tools across the enterprise.
- ROI on data initiatives (cost savings, revenue growth, enterprise value creation).

2025 market note

- Role has evolved from governance-first to value-creation-focused, with increasing emphasis on AI enablement and strategy.
- High demand in regulated industries (financial services, healthcare, manufacturing) and PE-backed growth companies.

Data Engineering Roles

DESIRED SKILL SETS: THE BASICS

In 2025, fundamentals of Python, SQL, Java, and Scala remain the foundation.

Cloud fluency in AWS, Azure, and GCP is now assumed, while distributed-framework expertise in Apache Spark, Kafka, and Databricks has become universal.

Familiarity with Pandas, NumPy, and dbt is table stakes, and employers increasingly expect Docker, Kubernetes, and workflow orchestration tools such as Airflow.

The market also emphasizes data governance, lineage, and observability, as organizations seek to ensure trust and compliance across expanding AI-powered data ecosystems.

DESIRED SKILL SETS: THE NEXT LEVEL

Organizations now prioritize engineers who can productionize AI pipelines — bridging data engineering with MLOps and DataOps, a core theme of this year’s report.

Experience with TensorFlow, PyTorch, and LLM integration is valued not for model training itself, but for enabling *model-ready data systems*.

Engineers capable of deploying CI/CD pipelines for data, integrating AI-assisted data-quality monitoring, and managing feature stores and metadata catalogs are commanding premium compensation and accelerated advancement.

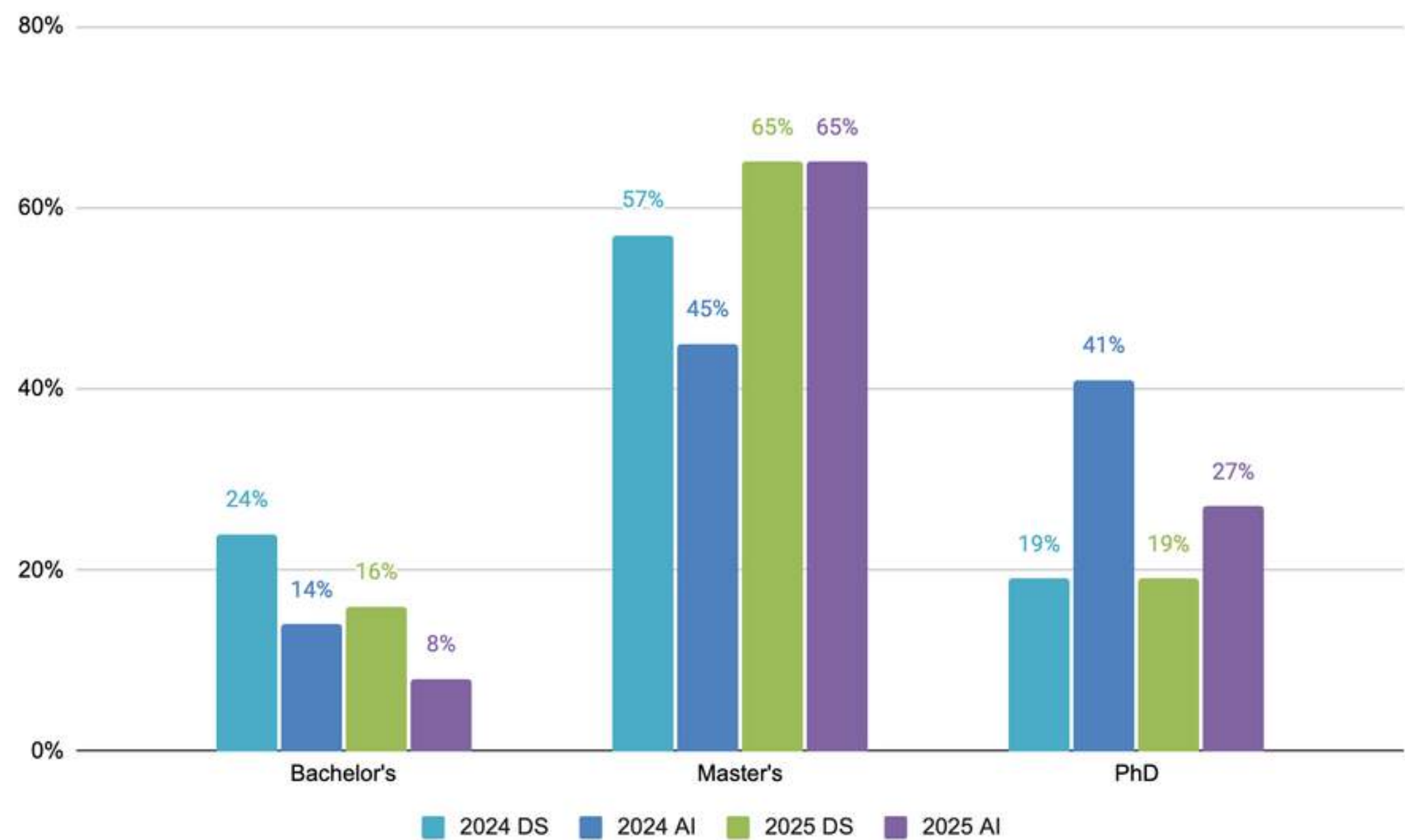


EDUCATION:

COMPARISON OF DEGREE LEVEL

- ~91% of AI professionals hold a graduate degree (Master's or PhD) in 2025 vs ~86% in 2024.
- Master's now dominates:
 - AI: Master's ≈ 64% in 2025 (up ~+19 pp YoY), overtaking PhD as the largest credential.
 - Master's has the sufficient expertise for using GenAI
 - Data Science: Master's ≈ 64% in 2025 (up ~+7 pp YoY).
- Bachelor's share keeps shrinking:
 - AI: ≈ 8% in 2025 (down from ~14%).
 - Data Science: ≈ 16% in 2025 (down from ~24%).
- Implication: Entry-level AI jobs increasingly demand a graduate credential or equivalent experience (bootcamps, high-profile open-source contributions). For DS roles, a Bachelor's remains viable, but the credential ceiling is rising.
- Implication: Unless you are working at the big labs or FAANG, a Master's degree is enough though a PHD degree may still be considered.

Comparison of Degree Level: Data Science vs AI (2024 vs 2025)



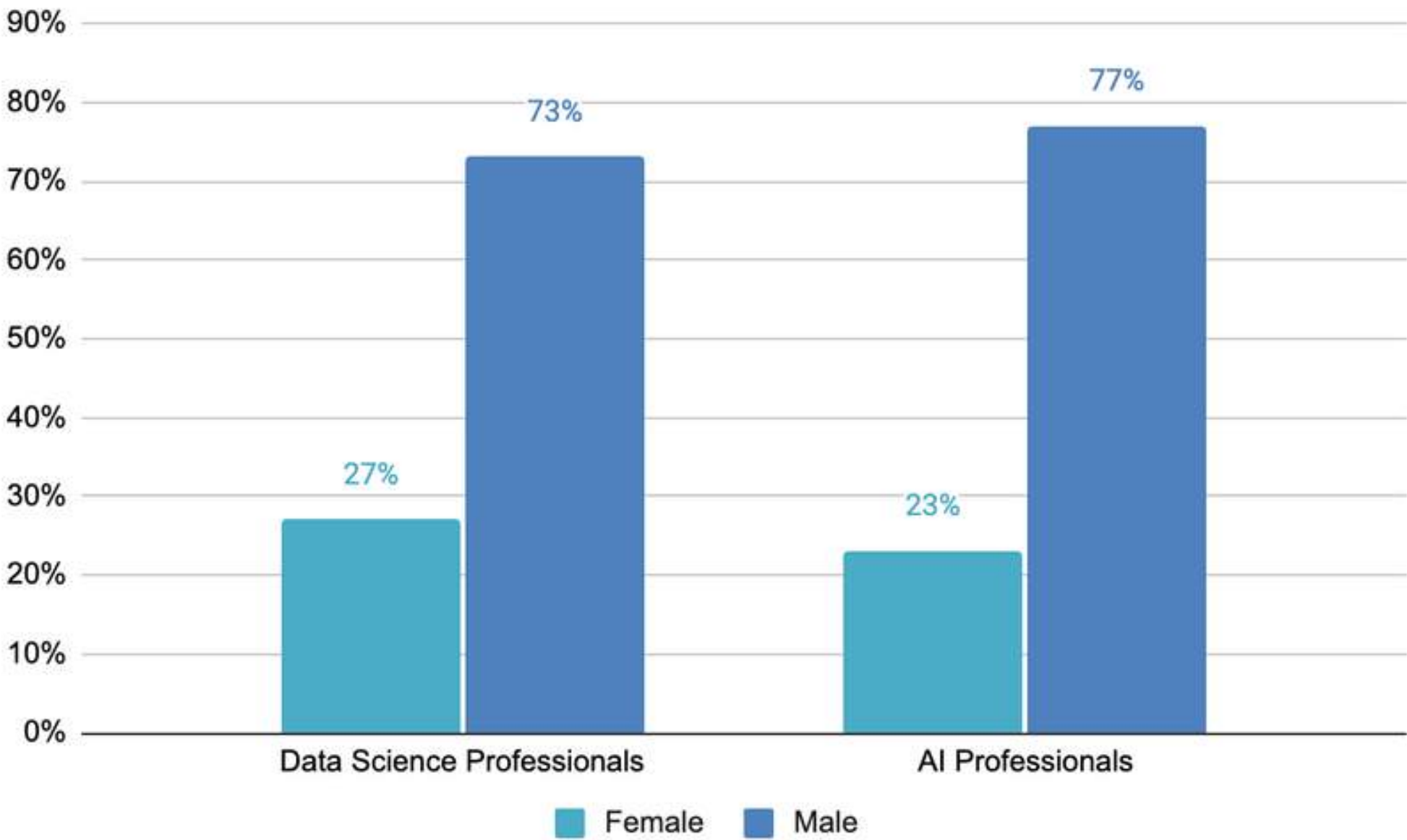
EDUCATION:
SALARIES BY JOB LEVEL FOR
DATA SCIENCE PROFESSIONALS

Job Level	Degree	25th	Mean	75th
IC-1	BS	\$72,500	\$94,643	\$110,000
	MS	\$88,750	\$110,027	\$130,000
	PhD	\$127,500	\$131,818	\$137,500
IC-2	BS	\$115,000	\$136,994	\$150,000
	MS	\$120,000	\$138,421	\$150,000
	PhD	\$130,000	\$148,146	\$160,000
IC-3	BS	\$130,000	\$150,888	\$170,000
	MS	\$140,000	\$159,073	\$177,500
	PhD	\$150,000	\$163,021	\$180,000
MG-1	BS	\$130,000	\$151,045	\$173,500
	MS	\$150,000	\$162,496	\$180,000
	PhD	\$175,000	\$183,333	\$200,000
MG-2	BS	\$163,775	\$190,156	\$200,000
	MS	\$170,100	\$194,030	\$206,500
	PhD	\$172,550	\$189,704	\$200,000
MG-3	BS	\$242,750	\$263,833	\$300,000
	MS	\$200,000	\$239,895	\$250,000
	PhD	\$200,000	\$252,871	\$293,750

EDUCATION:
SALARIES BY JOB LEVEL FOR
AI PROFESSIONALS

Job Level	Degree	25th	Mean	75th
IC-1	BS	\$146,250	\$152,500	\$158,750
	MS	\$110,000	\$130,063	\$150,000
	PhD	\$110,000	\$120,000	\$130,000
IC-2	BS	\$110,000	\$120,000	\$130,000
	MS	\$130,000	\$149,736	\$165,000
	PhD	\$120,050	\$145,743	\$165,000
IC-3	BS	\$140,000	\$173,040	\$185,000
	MS	\$160,000	\$165,623	\$180,000
	PhD	\$150,000	\$175,891	\$200,050
MG-1	MS	\$142,500	\$155,000	\$167,500
	PhD	\$147,500	\$165,000	\$182,500
MG-2	BS	\$215,050	\$223,367	\$235,000
	MS	\$177,525	\$204,192	\$212,500
	PhD	\$170,000	\$177,914	\$180,050
MG-3	BS	\$250,000	\$250,000	\$250,000
	MS	\$200,000	\$246,144	\$275,000
	PhD	\$225,000	\$241,667	\$272,500

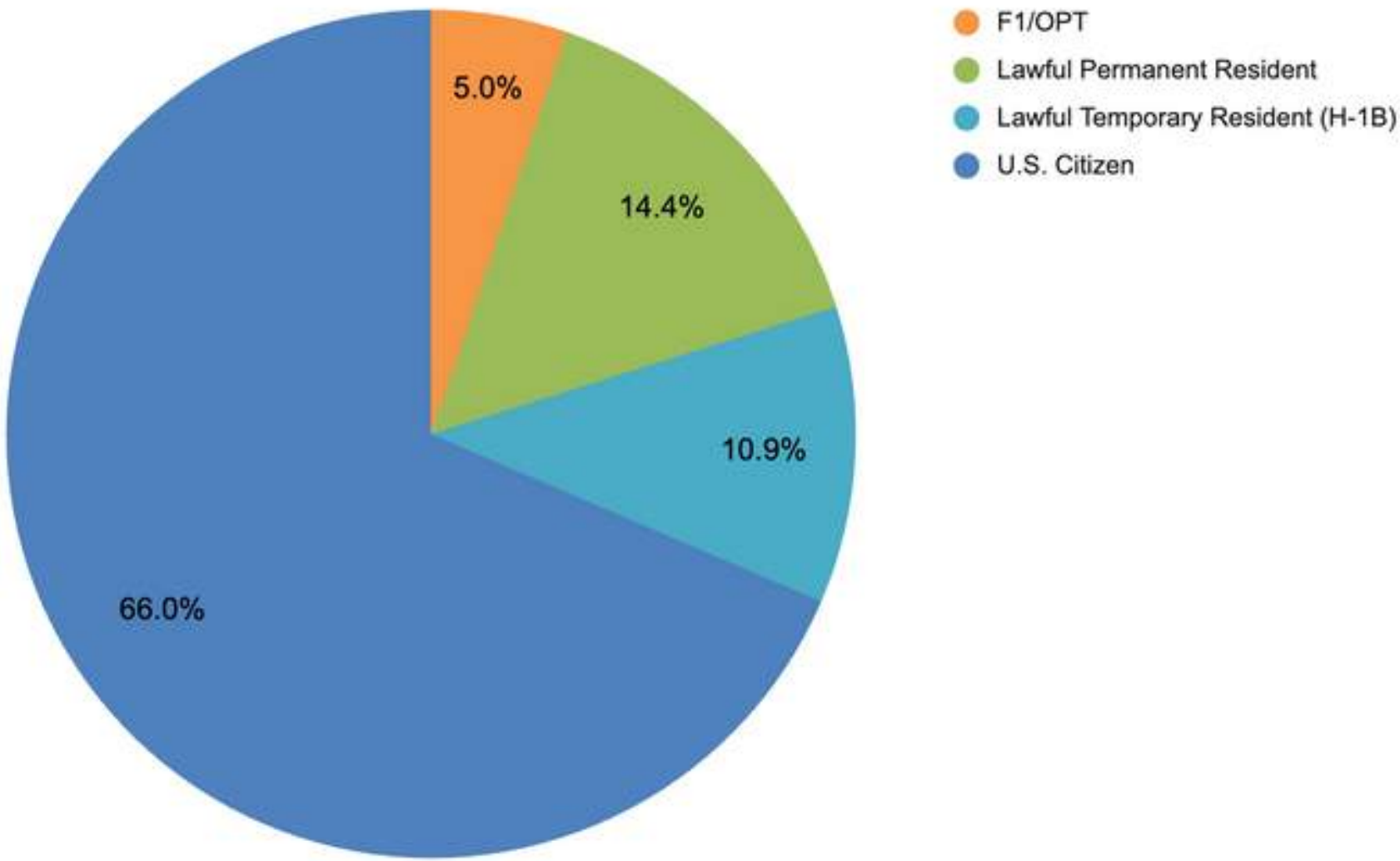
GENDER BREAKDOWN: DATA SCIENCE & AI



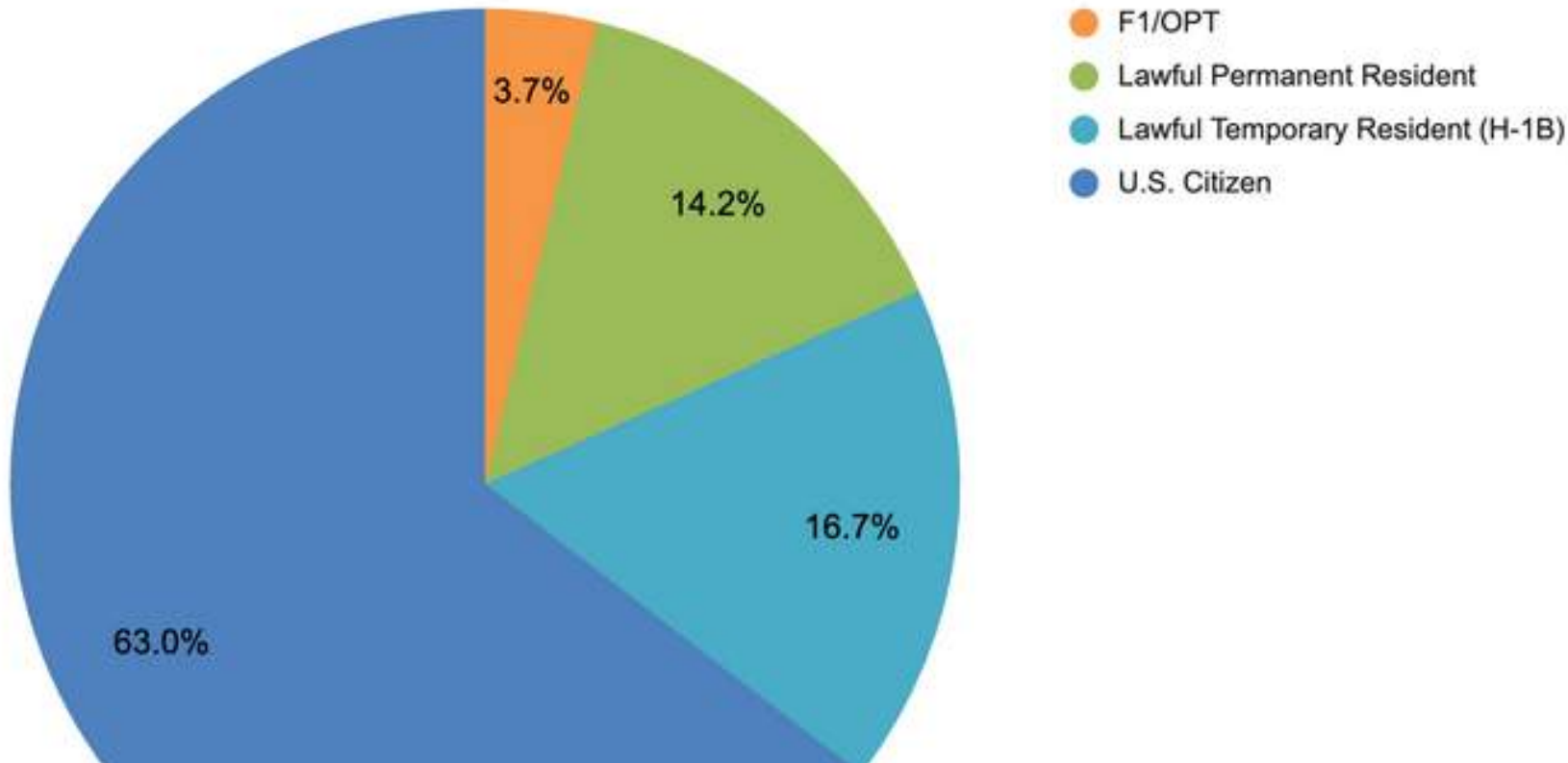
CANDIDATE RESIDENCY STATUS

Companies appear more willing to hire non-US citizens (F1/OPT, lawful permanent residents, and H-1B visa holders) for AI roles, compared to Data Science positions.

Data Science Professionals Residency Status Breakdown



AI Professionals Residency Status Breakdown



2025 Burtchworks Research Professionals Salary Report

Key Takeaways

2025: DATA INTEGRATION, RESEARCH, AND ADVANCED TOOLS

Employers and professionals are navigating an evolving landscape shaped by the following factors:

Integration of Data

Market researchers are increasingly expected to merge primary research with secondary sources and big data, enabling a more comprehensive consumer view.

Cross-disciplinary Collaboration

Collaboration with data science, analytics, and technology teams is becoming the norm, particularly for projects requiring advanced insights and predictive capabilities.

Adoption of Advanced Tools

There is a growing emphasis on tools like Python, R, SQL, and machine learning models to enable deeper insights and foster automation within research processes.

Increased Focus on ROI

Companies are emphasizing measurable outcomes, requiring market researchers to align insights with clear business objectives and demonstrate return on investment (ROI) for research initiatives.

Expansion of DIY Research Platforms

Tools like Qualtrics and SurveyMonkey are enabling organizations to conduct quick-turn studies, although this trend emphasizes the need for professionals to guide proper design and analysis.



Current Market Impacts

UPTICK IN HIRING AND HYBRID WORK MODELS

Job Market Flux

The market has shifted from an employer-driven hiring freeze in previous years to increased hiring activity in 2024. This uptick has generated optimism for sustained growth through 2025.

Hybrid Work Models

Hybrid working remains the most popular arrangement, with client-side roles typically requiring 2-3 days of in-person attendance. Supplier-side organizations continue to offer greater flexibility for remote work.

Competitive Landscape

While the candidate pool remains rich due to layoffs in 2023, companies are now actively seeking to retain top talent by streamlining hiring processes and offering attractive packages.



Navigating the Market

HOW TO NAVIGATE THE CURRENT HIRING MARKET



Growing Importance of Data Integration

Market researchers are now expected to combine primary research with secondary and big data sources, creating a more holistic view of consumers.

Cross-disciplinary Collaboration

Engagement with data science teams is becoming standard, so proficiency across these technical areas is a benefit for market research professionals.

Emerging Tools and Platforms

New tools and platforms are emerging for automation, visualization, or AI-driven analysis. Market researchers should adopt more advanced tools and techniques such as Python, R, and SQL, and learn machine learning models to derive deeper insights from data.

Supplier side:
Consultancy / X as a service

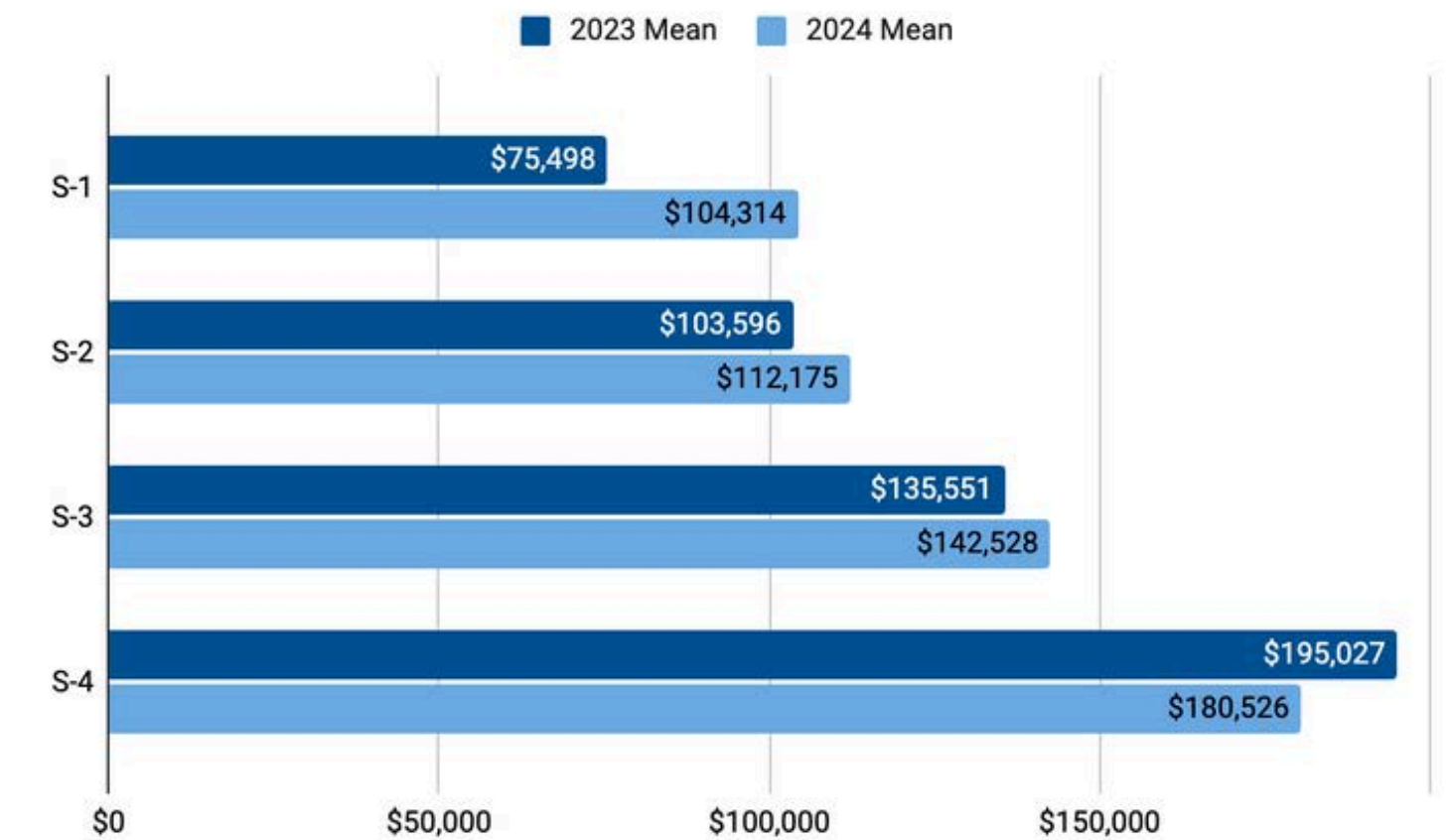
Client side:
In-house for a company

Supplier-Side	
Level	Typical Years of Experience
S-1	0-4 years
S-2	5-8 years
S-3	9-14 years
S-4	15+ years

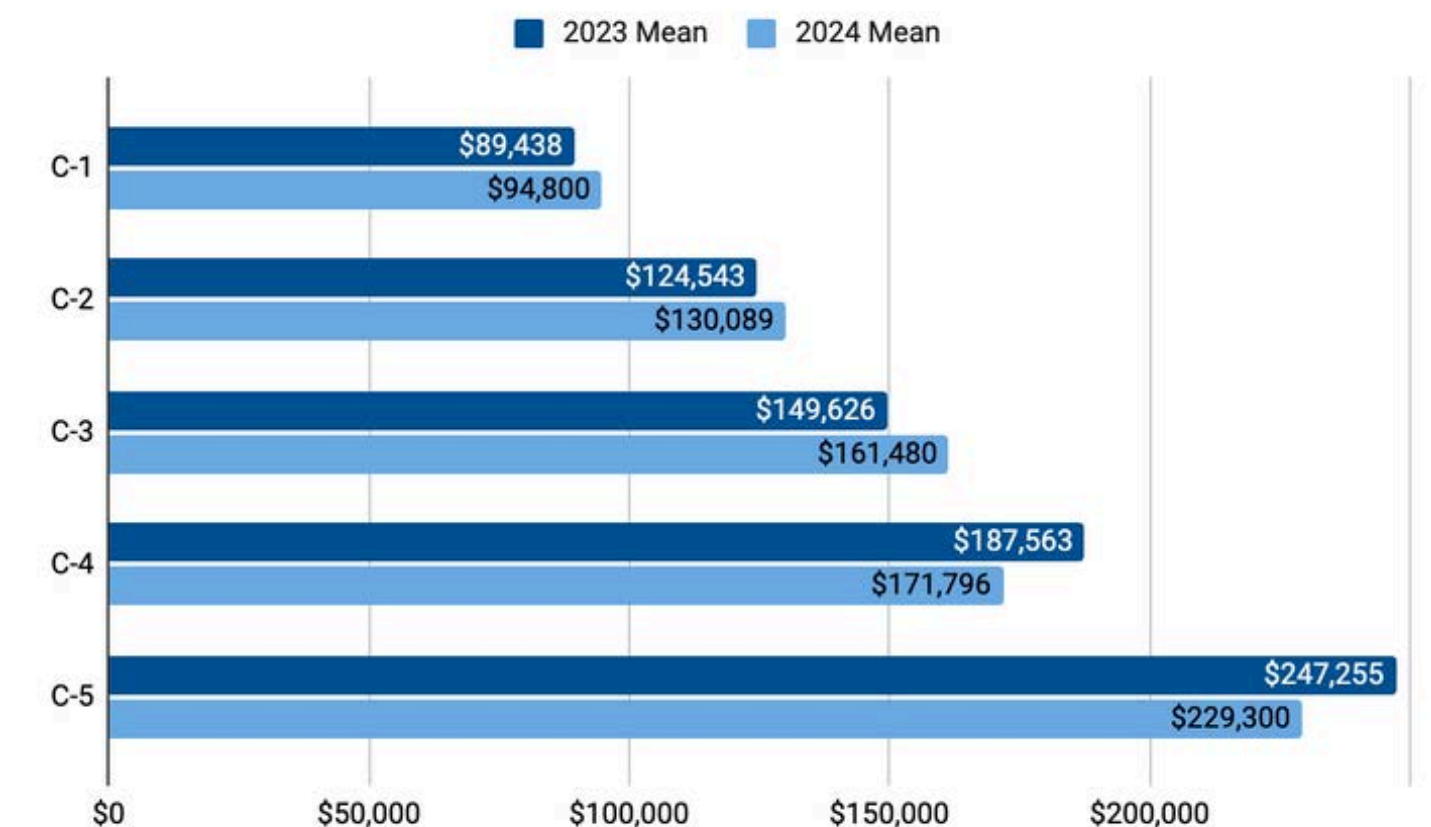
Client-Side	
Level	Typical Years of Experience
C-1	0-4 years
C-2	5-8 years
C-3	9-14 years
C-4	15-24 years
C-5	25+ years

COMPENSATION CHANGES: 2023 AND 2024 BASELINE DATA

Comparison of Supplier-Side Mean Base Salary: 2023 v. 2024



Comparison of Client-Side Mean Base Salary: 2023 v. 2024



**What else should I be thinking
about in my career?**

Soft skills

<https://www.manning.com/books/build-a-career-in-data-science>

<https://www.youtube.com/watch?v=pBisSm0Zr3A>

BUILD A CAREER IN Data Science

Emily Robinson
Jacqueline Nolis

 MANNING



**But lets end on a positive note
and advice for you....**

My advice...

- The world will turn, everything will be different in not too long
- Stay curious! Keep learning!
- Think about the big picture once in a while, including ethics
- Learn about the system you're part of, and try to change it
- Remember that there is no single definition of success. Your grades, job title, and bank account do not define you.
- Even as you strive to succeed, don't let it take over your life!
Stay happy and balanced!!

My hope for you...

- Many productive failures
- Beautiful triumphs to match the beautiful failures
- A sense of purpose in your life
- Many moments where you feel achievement and joy!

**Thanks to your
amazing
instruction team!**

Beijie Cheng

Chaitanya Kapoor

Jason Chen

Shreya Musini

Yueying Dong

Zhicheng Wang

And thanks to YOU!